

DOINGG@UNION.EDU

COURSE: CSC106 CAN COMPUTERS THINK?

SELF

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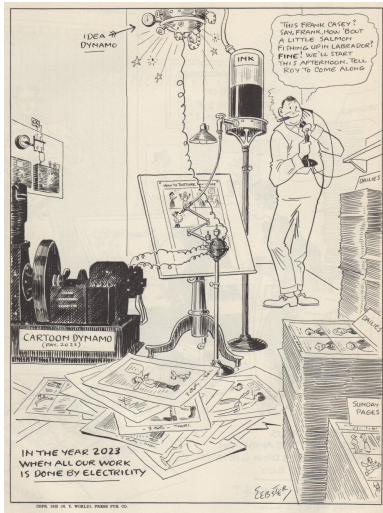
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Home



The course website for [CSC106](#), part of the Union College [CS curriculum](#). Here you can find our weekly schedule, assignments and resources.

Announcements

Welcome! You may want to bookmark this page for the term.

Office Hours

- Student/Office Hours:

- Steinmetz 108B
- Monday 2:00 pm – 3:00 pm
- Thursday 2:00 pm – 4:00 pm
- subject to change, check course website for most up-to-date schedule
- drop-in or schedule a 15 minute slot: <https://calendar.app.google/8bus6pfDvyphR9ar5>
- by appointment for another time or over zoom

Course Description

Can computers think? Is it possible to build intelligent machines? What does it mean for a machine to think or to be intelligent? Why, or why not, would we want intelligent machines? To what extent do we already have them? How do they work? How do they impact our lives? What are their limitations? This course invites you to explore these questions and more.

This course is an introduction to the field of computer science with an artificial intelligence theme that does not assume you have prior experience with computer programming or computer science. It introduces basic algorithms, data structures and programming techniques. You will learn how computer scientists think about and solve problems by doing your own computer programming. This will give you an understanding of how the current technology for building intelligent machines works. In addition, you will read non-technical papers about artificial intelligence and the ethical questions raised by its use. You will reflect on the ways current computational systems often reinforce biases and can harm groups of people inequitably. And you will evaluate how the choices you are making when creating your own programs will affect future users.

If you're thinking about a CS or CPE major or minor, this course will give you a solid foundation to continue to *Programming on Purpose* (CSC 120), to *Data Structures* (CSC 151) and beyond. If you're a neuroscience major, it will help you see how scientists are using biological and neurological principles to model "behavior" in computers and it will teach you skills useful for processing experimental data. If you are thinking about another major or minor, this course will give you foundations to apply computing in whatever area you are interested in. And if you're here just because you're curious, that's great!

Schedule

Weekly Schedule

Schedule

check often as things may change

W	TOPIC	Due	Runestone
1	Programming, Calling Functions	Introductory Survey	2.1 - 2.13
2	Loops, Variables and Assignment	HW 1	3.1-3.5
3	Defining Functions	HW 2, Project 1	5.1-5.12
4	Conditional Statements	HW 3, Practical Exam 1	4.1-4.9
5	Conditional Loops	HW 4, Project 2	6.1-6.7
6	Strings, Lists	HW 5, Written Exam 1	7.1-7.12
7	Nested Lists	HW 6, Project 3	9.1-9.14
8	File I/O, Dictionaries	HW 7, Practical Exam 2	8.1-8.9
9	Dictionaries	HW 8, Project 4	10.1-10.6
10	Searching, Sorting, Recursion	HW 9, Written Exam 2	

Resources

Runestone

- <https://runestone.academy/runestone/default/user/register>
- your Union College email
- unioncollege_py4e-int_spring26 as the course name

Gradescope

- <https://www.gradescope.com/courses/1288185>
- Entry Code: ZJ8ZRW

Python

- Python 3.13.X: the software package needed to write and run Python programs
- <https://www.anaconda.com/download/success?reg=skipped>
- Miniconda works just as great and takes up less space on your computer

Python Documentation

- Official Documentaion: <https://docs.python.org/3/index.html>
- Can *always* be used as a reference
- Can be searched for module-specific documentation e.g. turtle

Reeborg

- Reeborg's World: a useful website for learning some programming basics
- Reeborg's World: <https://reeborg.ca/reeborg.html>
- [Reeborg Project Documentation](#)

PythonTutor

- Python Tutor: a useful website for tracing python code
- <https://pythontutor.com/visualize.html#mode=edit>

Google Colab Pro

- Pro account sign-up: <https://colab.research.google.com/signup>
- Use your Union email
- A pro account is not needed for this course, just an option Union pays for
- Notes template: <https://colab.research.google.com/drive/15rPmKyQKCCbwsdDCpC39o8hczWZ-B5zV?usp=sharing>

Python Style Guide

- PEP 8
- should be used in all code cells of Colab notebooks
- Link: <https://peps.python.org/pep-0008/>

Syllabus

Syllabus

CSC106 Syllabus

1 CSC 106: Can Computers Think?

- Union College, Spring 2026
- Georgia Doing, PhD: email: doingg@union.edu, office: Steinmetz 108B

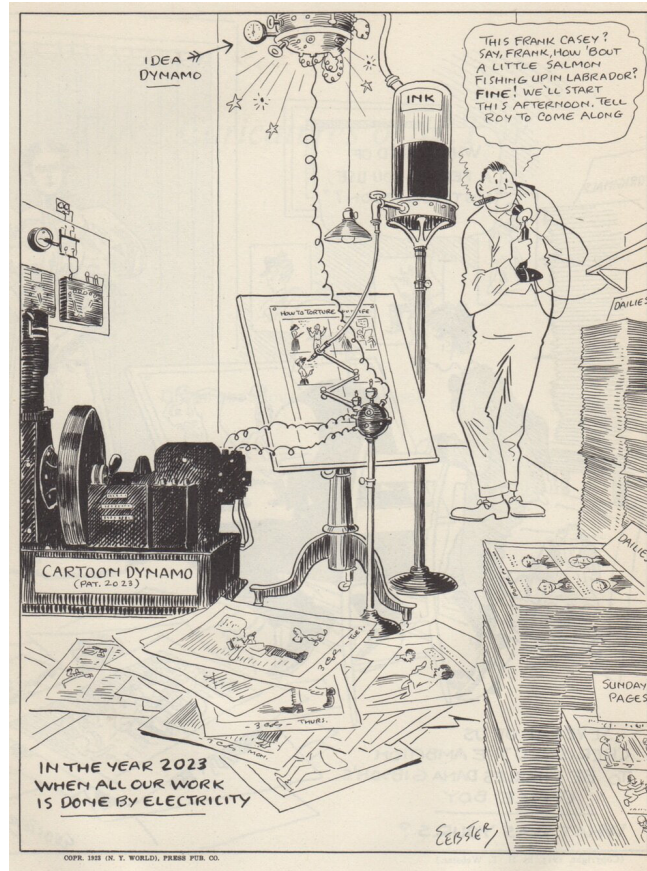


Figure 1: Cartoon by H.T. Webster published in 1922.

2 COURSE BASICS

- Lecture: Monday, Wednesday, Friday 1:50 pm - 2:55 pm
- Lab: Thursday 10:55 am - 12:40 pm
- Location: Olin 107
- Student/Office Hours:
 - Monday 3:00 pm - 4:00 pm
 - Thursday 2:00 pm - 4:00 pm
 - subject to change, check course website for most up-to-date schedule
 - drop-in or schedule a 15 minute slot: <https://calendar.app.google/gnYNtHjwwNxJ2UUN7>
 - by appointment for another time or over zoom

3 OVERVIEW

Can computers think? Is it possible to build intelligent machines? What does it mean for a machine to think or to be intelligent? Why, or why not, would we want intelligent machines? To what extent do we already have them? How do they work? How do they impact our lives? What are their limitations? This course invites you to explore these questions and more.

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If you're thinking about a CS or CPE major or minor, this course will give you a solid foundation to continue to *Programming on Purpose* (CSC 120), to *Data Structures* (CSC 151) and beyond. If you're a neuroscience major, it will help you see how scientists are using biological and neurological principles to model "behavior" in computers and it will teach you skills useful for processing experimental data. If you are thinking about another major or minor, this course will give you foundations to apply computing in whatever area you are interested in. And if you're here just because you're curious, that's great!

4 LEARNING OBJECTIVES

By the end of the course, you should be able to answer the following questions:

- What are the most important programming fundamentals and why are they the building blocks of programs?
- What are the debugging fundamentals and how do you use them?
- What are some of the mechanisms used to build “intelligent” machines?

By the end of the course, you should be able to do the following:

- Use variables, functions, conditionals, loops and recursion
- Read Python programs
- Write short Python programs that solve a given problem
- Fix short Python programs using debugging fundamentals

Computing/programming topics we will cover include functions, variables, assignment, conditionals, loops, recursion, lists/arrays, file reading/writing, binary numbers and ASCII, strings and string methods and dictionaries.

A 1-page schedule-at-a-glance is included at the end of the syllabus, but please see Nexus for a link to a more detailed schedule.

5 CLASS POLICIES AND GUIDELINES

The Computer Science Department as a whole welcomes all people, regardless of age, background, beliefs, ethnicity, gender identity, gender expression, national origin, religious affiliation, sexual orientation and any other differences, be they visible or non-visible. It's also important to recognize that institutional racism has prevented members of marginalized groups - especially black, indigenous and other people of color (BIPOC) - from fully participating in the field of Computer Science.

You. Yes you, the one reading this syllabus. You belong here.

As an instructor, I will do my utmost to uphold these principles and to treat everyone with respect. As students in this class, I expect you to do the same since one person is not a community unto themselves. We're all in this together, so let's treat each other well. Lastly, I welcome feedback on issues we might discuss or ways that our class can be a more just one for all people.

6 COURSE MATERIALS

Online Textbook: Python for Everybody - Interactive (free).

We are going to use an online, interactive, free and open source textbook. This textbook has a lot of embedded interactive exercises. Reading this book always also means doing the activities. To get access to the book, please register on Runestone using the following information which can also be found on Nexus:

- <https://runestone.academy/runestone/default/user/register>
- your Union College email
- unioncollege_py4e-int_spring26 as the course name

We'll use the following websites and software throughout the course:

- The course website: a github-hosted site with assignments, materials and announcements
 - <https://georgiadoing.github.io/CSC106-S26/>
 - you may want to bookmark this to check it before and after each class
- Gradescope: used to submit programming assignments
 - <https://www.gradescope.com/courses/1288185>
 - Entry Code: ZJ8ZRW
- Python 3.13.X: the software package needed to write and run Python programs
- Reeborg's World: a useful website for learning some programming basics
- https://reeborg.ca/index_en.html
- Python Tutor: a useful website for tracing python code
 - <https://pythontutor.com/visualize.html#mode=edit>

7 COURSE RESOURCES

The course website has a list and links to the readings and software resources that we are going to use. In class, we are going to use the Linux machines in Olin 107. Outside of class you have a choice of hardware. You can install the software on your own computer (Python is freely available for Windows, Mac, and Linux, Reeborg's World and Python Tutor are browser based programs) or you can work in one of the Computer Science labs. We have three spaces that you can use 24/7 using your ID card, except when classes are being held in them:

- Olin 107 - where we have class
- PASTA Lab (ISEC 051+)
- Computer Science Resource room (Steinmetz Hall, 209A)

8 LIBRARY RESOURCES

The library is available to help you with your research needs! Librarians can help you develop research questions, search for and select the best sources for your projects, identify research strategies, evaluate sources, and assist you with creating citations. There are multiple ways for you to contact a librarian. For more information, please see our Ask A Librarian page: <https://www.union.edu/schaffer-library/ask-a-librarian>.

9 ACCOMODATIONS

It is the policy of Union College to make reasonable accommodations for qualified individuals with disabilities. If you are a person with a disability and wish to request accommodations to complete your course requirements, please make an appointment with me or stop by during my office hours as soon as possible, all discussions will remain confidential. You must provide reasonable notice and be in touch with Accomodative Services on the 2nd floor of Schaffer Library, if you have not already, if you wish to take advantage of extra time on exams.

10 COMPLEX QUESTIONS: GLOBAL CHALLENGES & SOCIAL JUSTICE

Justice, Equity, Identity, Difference (JEID). Algorithms and computational systems increasingly shape the world we live in. Advances in AI and machine learning are in the news on a daily basis. But there is also a growing awareness of the limitations and dangers of these systems. In particular, they often promote biases and inequalities. For example, facial recognition software has been shown to work much less accurately for people of color than for white people, an AI hiring algorithm was found to reject female applicants for technical positions at a higher rate than male applicants, and search results can be full of racial stereotypes. In this course, you will learn about real-world cases of AI systems that discriminate against groups of people. You will see that related ethical questions are raised even in the context of the relatively simple programs you create in an introductory computer science class. Additionally, you will work on developing a habit of always evaluating how your own choices as a programmer might impact other people's experiences.

Engineering, Technology and Society (ETS). In this course, you learn to break down a larger problem into a series of smaller computational steps (creating an algorithm) and to implement algorithms in Python. Through these activities you will gain an understanding of how computers work and an idea of the software development process. In particular, you will learn about iterative development and the importance of thorough testing. You will practice communicating about algorithms and their implementation in groups.

You will also explore, through assignments and discussion, how computational systems can have unintended and negative consequences for groups of people and what the responsibility of the programmer is to mitigate these dangers.

Data & Quantitative Reasoning (DQR). Much of current AI systems are data driven and learn based on collections of data. In this course, you will learn about and implement some approaches that AI systems use to draw inferences based on data. Many of the ethical issues around current AI systems are linked to the data these systems use. Throughout the course, you will, therefore, also reflect on: What information is collected? How is the information represented? Who is represented in the data? How is it used?

11 GRADING

The following components will contribute to your final grade for this class with roughly the indicated weight.

- 2 practical exams: 20% (10% each)
- Written midterm: 15%
- Written final exam: 15%
- Programming projects: 40%
- Engagement: 10%

Note: You must get a C- or better in order to take any other class that has a CSC-10x prerequisite. Several larger programming projects will be assigned to reinforce the concepts discussed in class and put into practice in homework assignments. All projects can be completed using programming techniques that have been covered in class.

Do not use techniques that have not been covered in class unless specifically told otherwise. Part of the challenge for each programming project is figuring out how to complete the assignment using only the tools provided in the course up to that point.

Letter Grade Scale:

- A: 93-100; A-: 90-92
- B+: 87-89; B: 83-86; B-: 80-82
- C+: 77-79; C: 73-76; C-: 70-72
- D: 60-69; F: 0-59

12 ATTENDANCE

Class participation is a critical component of the course and attendance is mandatory. I realize that sometimes other things come up (interviews, athletic events, illness, emergencies etc.).

In those cases, just let me know (in advance, if at all possible) that you are going to be absent. There may not always be the possibility of make-up credit for missed in-class material, but it is more likely for something to be arranged if you discuss your absence with me prior to class. No matter how much class you have missed, please do not come to class if you have an illness that is likely contagious, please email me to work out a reasonable solution.

If you do miss class, it is your responsibility to make up any material that you missed. Get notes from a classmate, make sure to complete all homework assignments due or assigned during the class that you missed, and come see me if you have any questions on the material. I will expect you to hand in all assignments by the due date, regardless of whether you were in class. You will not be able to make up missed exams, quizzes or in-class engagement points.

13 LATE POLICIES

Homework assignments will usually be discussed in class on the day that they are due and will not be accepted late. Programming projects are due by 11:59:59pm on the due date. You have three “extension tokens” for this class. Each token will extend a single project deadline by 24 hours. You can use each token on a different project, or use two or even all three on a single project. Once all three tokens have been used, no late submissions will be accepted (barring exceptional circumstances).

14 PARTICIPATION AND ENGAGEMENT

Following these guidelines will constitute positive participation and engagement and earn you up to a 0.5% of your final grade per week:

- **Respect.** This course is a space for rigorous and respectful debate. When confronting ideas and people different from you, lead with curiosity rather than judgement. This commitment extends to all our face to face and digital interactions. We will critique ideas, not people. To protect our shared learning environment, you may not record, photograph, or share any part of our class sessions outside of our class.
- **Engage.** Show up to class on time and prepared. Show up ready to learn by completing the readings and exercises and checking the course website often. When you are late or unprepared, you are disrespecting the learning experience for the group. Engagement in class constitutes being present, respectful and engaged. Accessing off-topic websites or software, checking email or phones, *utilizing large language models* or otherwise attending to things not pertinent to the course is distracting to yourself and others.

- Embrace intellectual humility. Recognize that there are limitations to your knowledge and that some of your beliefs could actually be wrong. Be curious about your thinking and open to learning from others. This is hard, but so vital to your learning in this course—we'll work on developing this throughout the term!
- Get help when you need it. If you are stuck or confused or lost, be proactive and get help. The best learners and thinkers can figure out when they are stuck and make a plan to get unstuck. Come to Office Hours (Steinmetz 108B), the CS Helpdesk (Sun-Thurs 7:00-9:00pm Olin 107), Biology Backup (ISEC 316, Tues & Thurs 5:30-7:30pm) or ask another student. Often the best person to explain something is the person who just figured it out. Try reaching out to another student in the course who seems like they get it—they will probably be flattered you asked!

15 ACADEMIC INTEGRITY AND "ARTIFICIAL INTELLIGENCE" USE

Union College recognizes the need to create an environment of mutual trust as part of its educational mission. Responsible participation in an academic community requires respect for and acknowledgement of the thoughts and work of others, whether expressed in the present or in some distant time and place. Matriculation at the College is taken to signify implicit agreement with the Academic Honor Code, available at: <http://muse.union.edu/honorcode>

It is each student's responsibility to ensure that submitted work is their own and does not involve any form of academic misconduct. Students are expected to ask their course instructors for clarification regarding, but not limited to, collaboration, citations, and plagiarism. Ignorance is not an excuse for breaching academic integrity. Students are also required to affix the full Honor Code Affirmation, or the following shortened version, on each item of coursework submitted for grading:

'I affirm that I have carried out my academic endeavors with full academic honesty.'

[Signed, Jane /John /Jaden Doe]

What it means for this course: **you must complete the programming projects on your own, but you are welcome and encouraged to collaborate on other regular assignments.**

Please notice that different rules apply to the exercises that are part of the textbook, regular assignments and programming projects.

Textbook Readings and Regular Exercises:

You are welcome to discuss the textbook readings, the exercises embedded in the textbook, and other regular assignments with other people in the class to help you fully understand the material. However, don't just copy somebody else's solution. Even if you ask other people for help, the goal is for you to understand the underlying concepts and logic. It's more important to understand how to arrive at a correct solution than it is to know the solution itself.

Programming Projects:

Any work you turn in has to be your own. The best way to learn programming is by doing programming. I expect that you already know that if you do not do things for yourself, you will not learn them. Sometimes that will mean that you will have to struggle. That is ok: struggling to figure out how to solve a problem is where a lot of learning happens. Give yourself the opportunity to do this.

Please, do not ever copy any part of a solution from a friend, from any material generated by an “artificial intelligence” (AI) chatbot model (ChatGPT, Gemini, Copilot, Claude, etc.) or from anywhere else. Do not ever *look at* somebody else’s solution or any solution generated by an entity other than yourself before you have completed your own. Do not ever *show* somebody else your solution before you’ve both submitted it; though it may be beneficial to discuss your solution with another student in the class after you’ve both handed in the assignment. Please do not ever show your solution to future students of this class or any chatbot, and do not ever post your solutions on the Internet.

Don’t search for information on the Internet. Don’t ask for information from any AI model. Especially, don’t look at any material that provides solutions to exercises that are similar to the projects. An exception is, of course, any online material that is linked directly from the Nexus page. As always, you should practice good citation behavior and cite any sources that you consult in your work (for example, which pages from the Python documentation website you consulted when working on the project).

Discussing your work (without looking at code): Talking through a problem is extremely beneficial in understanding the problem and coming up with a solution path. So, I strongly encourage you to talk to other students in this class about the projects, visit the CS Helpdesk and come to my Office Hours (you can also request a meeting if you have schedule conflicts with my Office Hours).

Here are some ways in which you can talk about the projects without having to worry about violating the rules from above.

- It’s ok (and strongly encouraged!) to draw **memory diagrams** together. Draw memory diagrams and discuss possible algorithms for solving the problem.
- It’s ok to **write down an algorithm in English** together. Then go off by yourself to implement it into code.
- It’s ok to read and write code together that is not part of the assignment, but that helps demonstrate the concept of what you’re being asked to do. Go through an example from class or from the book together, or **come up with your own examples**. There’s no better way to understand something than trying to think up examples in order to teach it to someone else. Try it!

Do not work on the implementation together with friends, especially if you are each at your own computer and are always in lock step trying to figure out the same line or section of

code at the same time. Have a higher-level discussion (using one of the strategies suggested above), then work on the implementation on your own.

Your goal for discussions about any assignment should be that you come away with a better understanding of the problem and of possible ways to approach it so that you can then try out these approaches on your own. You should never leave a discussion with just an answer, without an understanding of how to arrive at that answer. A good general guideline for any discussion, or interaction with an AI model, is that you should not leave the discussion with anything written or typed.

16 CONTACTING ME OUTSIDE OF CLASS

The best method for contacting me outside of class is to stop by my office during Office Hours or whenever my office door is open. You can also schedule a time with me if my regular office hours don't work. Please contact me via email (doingg@union.edu) and we can arrange a phone or zoom call. I respond to emails as soon as possible, but it may take up to one day to get a response during the term, and longer between terms.

17 ADDITIONAL RESOURCES

Mental Health and Campus Resources

Union College is committed to supporting and advancing the mental health and well-being of our students. During the course of their academic careers, students often experience personal challenges that contribute to barriers in learning, such as drug/alcohol problems, strained relationships, chronic worrying, persistent sadness or loss of interest in enjoyable activities, family conflict, grief and loss, domestic violence, difficulty concentrating, problems with organization, procrastination and/or lack of motivation. Students also sometimes come to college with a history of learning difficulties (e.g., any form of special education), experience difficulties succeeding in a particular subject (e.g., math, reading), or have experienced some form of trauma be it emotional or physical (e.g., head injury). These mental health concerns can lead to diminished academic performance and can interfere with daily life activities. If you or someone you know has a history of mental health concerns or if you are unsure and would like a consultation, a variety of confidential services are available. The Eppler-Wolff Counseling Center provides free counseling and therapy services (including psychiatry) to all enrolled students. Please call (518) 388-6161 or visit the Wicker Wellness Center in person any weekday between 8:30 and 5:00 to schedule an initial contact appointment. Visit the Counseling Center website for more information.

In a crisis situation, or after hours, contact Campus Safety at (518) 388-6911. The National Suicide Prevention hotline also offers a 24-hour hotline at (800) 273-8255.

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students (dos_office@union.edu) for support or drop into office hours Monday - Friday 8:30 am - 5:00 pm in the Reamer Campus Center room 306. Furthermore, please notify me if you are comfortable doing so and I will provide any resources I can. The Union College Persistence Fund can provide financial support in times of unexpected hardship to cover needs such as emergency medical expenses not covered by insurance and basic living expenses. Additional sources of support are outlined on the Dean of Students webpage: <https://www.union.edu/dean-students>.

18 TENTATIVE SCHEDULE

W	TOPIC	Due	Runestone
1	Programming, Calling Functions	Introductory Survey	2.1 - 2.13
2	Loops, Variables and Assignment	HW 1	3.1-3.5
3	Defining Functions	HW 2, Project 1	5.1-5.12
4	Conditional Statements	HW 3, Practical Exam 1	4.1-4.9
5	Conditional Loops	HW 4, Project 2	6.1-6.7
6	Strings, Lists	HW 5, Written Exam 1	7.1-7.12
7	Nested Lists	HW 6, Project 3	9.1-9.14
8	File I/O, Dictionaries	HW 7, Practical Exam 2	8.1-8.9
9	Dictionaries	HW 8, Project 4	10.1-10.6
10	Searching, Sorting, Recursion	HW 9, Written Exam 2	

18.1 ADDENDA

Any community agreed upon additions:

19 PROGRESS TRACKER

	Week	Week	Week	Week	Week	Week	Week	Week	Week	Week
Learning Goals	1	2	3	4	5	6	7	8	9	10
What are the most important programming fundamentals and why are they the building blocks of programs?										
What are the debugging fundamentals and how do you use them?										
What are some of the mechanisms used to build "intelligent" machines?										
Use variables, functions, conditionals, loops and recursion.										
Write short Python programs that solve a given problem.										
Fix short Python programs using debugging fundamentals.										

Grading

Table 2: Grading Scheme

Course Element	Grade Point Contribution	Notes
Engagement	10%	see syllabus for details
Programming Projects	40%	4 at 10% each
Midterm	15%	written
Practical Exams	20%	coding
Final Exam	15%	written
total	100%	

Table 3: Letter Grade Scale

Letter Grade	Percent range
A	93 - 100
A-	90 - 92
B+	87 - 89
B	83 - 86
B-	80 - 82
C+	77 - 79
C	73 - 76
C-	70 - 72
D	60 - 69
F	0 - 59

Note: You must get a C- or better in order to take any other class that has a CSC-10x prerequisite. Several larger programming projects will be assigned to reinforce the concepts discussed in class and put into practice in homework assignments. All projects can be completed using programming techniques that have been covered in class.

Do not use techniques that have not been covered in class unless specifically told otherwise. Part of the challenge for each programming project is figuring out how to complete the assignment using only the tools provided in the course up to that point.

Assignments

HW 01

CSC106: Assignment 1

- due Monday 4/6/26 by class time (1:50 pm)
- in-class notes from Friday: https://drive.google.com/file/d/1UW3fMha6Ki4dDR_sidpeiV-PDeBSFEpi/view?usp=sharing
 - examples of comments, importing turtle, variables and for loops
 - also includes link to turtle library documentation

Part 1

- [Intro Survey](#)
- Runestone Chapters 1-3 (we'll discuss in class next week starting Monday)

Part 2

- Practice Problems for Week 1: [HW01 instructions](#)
- Starter code: [winding_sandbox.json](#)
- Submit code on Gradescope, bring written answers to class
- Graded on effort/completion

HW 02

CSC106: Assignment 2

- due Monday 4/13/26 by class time (1:50 pm)
- Runestone Chapter 5 (see schedule for specific sub-sections)
- Practice Problems for Week 2:
 - Submit code on Gradescope,
 - Bring written answers to class
 - Graded on effort/completion

NAME:

Practice Problems for Week 2

Please complete, bring to class (written answers) and submit on gradescope (code) by end of day on **Monday 4/13/26**.

You will be submitting the following files:

- square_chain.py
- retirement.py
- All the programming files **MUST** be named as instructed

Programming Part 1: Turtle Stairs

1. Create a file called square_chain.py for your code.
2. Write a Python program using turtle to draw something that draws a diagonal chain of 10 squares:

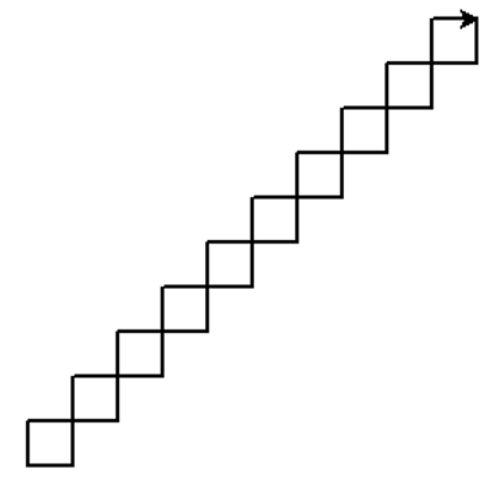


Figure 1: Turtle Stairs

There are many different ways (algorithms) that you can draw this pattern. Some may (or may not) leave the turtle in the same position and orientation that you see here. It does not matter if the turtle has the same position and orientation as what you see here.

3. Add comments to explain how your program works.
 - Header comment with Name, Email, Date, Assignment, Description
 - In-line comments describing the algorithm guiding your program
 - Docstrings for any functions, if you define new functions

Programming Part 2: Retirement Calculator

It's never too early to start saving for retirement. For this assignment you're going to write a simple retirement calculator.

- A run of the program may look like the following (user input in bold):

```
How much do you plan to save each month? (in dollars) **100**
How old are you? (in years) **20**
At what age do you plan to retire? (in years) **40**
What yearly rate of return do you expect? (in percentage points) **5**
Your savings at age 21 are: $ 1200
Your savings at age 22 are: $ 2460
Your savings at age 23 are: $ 3783
Your savings at age 24 are: $ 5172
Your savings at age 25 are: $ 6630
Your savings at age 26 are: $ 8162
Your savings at age 27 are: $ 9770
Your savings at age 28 are: $ 11458
Your savings at age 29 are: $ 13231
Your savings at age 30 are: $ 15093
Your savings at age 31 are: $ 17048
Your savings at age 32 are: $ 19100
Your savings at age 33 are: $ 21255
Your savings at age 34 are: $ 23518
Your savings at age 35 are: $ 25894
Your savings at age 36 are: $ 28388
Your savings at age 37 are: $ 31008
Your savings at age 38 are: $ 33758
Your savings at age 39 are: $ 36646
Your savings at age 40 are: $ 39679
```

1. Start by creating a file called retirement.py for your code.
2. Next, start writing code to get input from the user and store the information in variables.
 - The program should start by asking the user for the following information:
 - The amount they plan to save each month
 - Their starting age
 - Their planned retirement age
 - The yearly rate of return they expect (i.e. growth through interest or dividends).
3. Now design an algorithm to compute the amount that accumulates each year. Write out the algorithm in a human language (e.g. English) before writing it in code. Include this as your description in your header comment.
 - The program should show how the user's savings grow over the course of their careers. For example, if a 20 year-old starts saving \$100 a month, retires at 40, and gets a 5% rate or return, then they should have about \$39,679 by the time they retire.

- To simplify things, we assume that interest is added once per year before the user has contributed any of their monthly savings.
 - This means that, for the first year, the rate of return is zero, so they only get what they saved over the course of the year (i.e. \$1200).
 - For subsequent years, calculate the interest gained, then add the yearly contribution. So for year 2:
 - * the interest will be $\$1200 \times 5\% = \60
 - * the savings at the end of year will be $\$1200 + \$60 + \$1200 = \2460 .
4. Implement the program in Python
 5. Test the program with different values to ensure that it works correctly.
 6. If you haven't already, add comments explaining how the program works.

Written Questions**1. Consider the following (incomplete) Python program:**

```
x = int(input("Please input a positive integer: "))
counter = x
for step in range(0, 10):
```

Complete the missing lines of code above so that the program will produce the following output if the user inputs 3.

```
3
6
9
12
15
18
21
24
27
30
```

- **Notes:** You do not need to introduce any additional variables. There is more than one way to achieve this result.

2. Consider the following (incomplete) Python program to list the first few Fibonacci numbers:

The Fibonacci numbers are the following sequence of numbers:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, ...

- The first two numbers are 0 and 1.
- After that, each number can be calculated by adding the previous two together. For example:
 - the third number is 1 (0+1)
 - the fourth number is 2 (1+1)
 - the fifth number is 3 (2+1), etc.

They are named Fibonacci numbers after the Italian mathematician who first described them in Europe. Patterns based on the Fibonacci sequence show up in surprisingly many contexts in math, biology, and art. (See the Wikipedia entry if you want to know more.)

Below is the (incomplete) Python program to list the first few Fibonacci numbers.

The user gets to determine how many Fibonacci numbers to list.


```
1. print ("How many Fibonacci numbers would you like to see?")
2. limit = int (input("Please input a number greater than 2: "))
3.
4. fib_0 = 0
5. fib_1 = 1
6.
7. print(fib_0)
8. print(fib_1)
9. for count in range(      ):
10.     next_fib =
11.     print(next_fib)
```

- For questions a-d, assume that the program has been completed and functions correctly.

2.a. Draw a memory diagram that shows the - associations after line 5 has been executed. Assume that the user inputted 10. Also show what will have been printed to the screen at that point.

2.b. Draw a memory diagram that shows the - associations that should exist after line 11 has been executed for the first time. Also show what line 11 will print to the screen when it is executed for the first time.

2.c. What should the - associations be when line 11 has been executed for the second time? Draw a memory diagram.

2.d. Draw a memory diagram that shows the - associations after the last number has been printed (i.e. at the end of the program).

2.e. Complete the code above so that the program prints the first n Fibonacci numbers, where n is determined by the user. You may need to add new lines of code.

3. Consider two Python programs:

- Both programs on the next page will help Reeborg harvest the carrots in the Harvest 1 world:

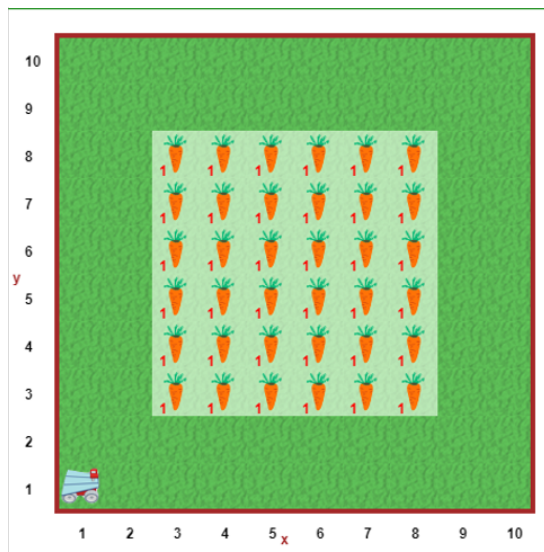


Figure 2: Reeborg Carrots

- Both programs define a number of functions. Unfortunately, the program designers were from the bad old days when names needed to be as short as possible, so their function names are essentially meaningless. Come up with better names for each of the functions that they defined.

-
-
-
-
-
-
-
-

Program 1

```
1 def turn_right():
2     turn_left()
3     turn_left()
4     turn_left()
5
6 def a():
7     move()
8     turn_left()
9     move()
10    move()
11    turn_right()
12    move()
13
14 def b():
15     for carrot in range(0, 6):
16         take()
17         move()
18
19 def c():
20     turn_left()
21     move()
22     turn_left()
23     move()
24
25 def d():
26     turn_right()
27     move()
28     turn_right()
29     move()
30
31 a()
32 for double_row in range(0, 3):
33     b()
34     c()
35     b()
36     d()
```

Program 2

```
1 def turn_right():
2     turn_left()
3     turn_left()
4     turn_left()
5
6 def e():
7     move()
8     turn_left()
9     move()
10    move()
11    turn_right()
12    move()
13    take()
14
15 def f():
16     for carrot in range(0, 5):
17         move()
18         take()
19
20 def g():
21     f()
22     turn_left()
23     move()
24     turn_left()
25     take()
26
27 def h():
28     f()
29     turn_right()
30     move()
31     turn_right()
32     take()
33
34 e()
35 for double_row in range(0, 2):
36     g()
37     h()
38     g()
39     f()
```

3.i Now that you've come up with better names for the functions, which of the solutions do you think is better and why?

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4. Check-in

4.a. Were you able to complete all of the programming activities (yes/no)?

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4.b. What, if anything, did you struggle with when working on the activities?

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.....

4.c. If you managed to overcome the challenges, how did you do it?

.....

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Weekly Notes

Week 1

Notes from Week 1

- Day 1 (Mon): [Seymour Pappert Turtle Documentary](#)
- Day 2 (Wed): [Instructions for First Program Activity](#)
- Day 3 — Lab (Thurs): [Reeborg activity: newspaper](#)

Python Examples

- in-class examples from Friday: https://drive.google.com/file/d/1UW3fMha6Ki4dDR_sidpeiV-PDeBSFEpi/view?usp=sharing
 - examples of comments, importing turtle, variables and for loops
 - also includes link to turtle library documentation

Week 2

In-class activities for week 3

- Mon: [Runestone Ch. 1 review](#)
- Wed: [More Turtle Drawing](#)
- Thurs (lab): [Variables and Loops](#)
- Fri: [Tracing & Memory Diagrams](#)

Key Resources this week

- all also found in [Resources page](#)
- [Python Documentation](#)
- [Python Tutor](#)

About

References

